

IBEH CHIMCHETAM CHRISTABEL/ HEALTH INDUSTRY

Massachusetts General Hospital: Performance & Insights Report

Dataset Used: Massachusetts General Hospital Patient, Encounter, Payer and Procedure Records.

Tool(s) Used: Excel

Number of Dashboards/Pages Created: Three Interactive Dashboards (Three Pages)

Industry: Health

STC Data Analyst Project - Week 4 Report

Massachusetts General Hospital: Performance & Insights Report

1. Summary

Massachusetts General Hospital is currently facing a systemic challenge characterized by a **60.18% 30-day readmission rate**—significantly higher than the national benchmark of ~15%. While the **Ambulatory** department handles the highest volume of patients, the **Inpatient** and **Urgent Care** departments drive the highest costs and operational bottlenecks. Addressing data quality issues regarding Length of Stay (LOS) and implementing targeted discharge protocols are critical for institutional stability.

2. Stakeholder-Specific Findings & Recommendations

Hospital CEO:

- **Readmission Crisis:** The 30-day readmission rate stands at **60.18%**. Historical data suggests this is a systemic pattern rather than a temporary spike, with rates consistently staying above 59% since a peak in 2014.
- **Operational Volume:** The hospital managed **27,891 total encounters**, with the Ambulatory department (12,537) and Outpatient (6,300) being the primary drivers of patient volume.

Recommendation:

Establish monthly tracking of headline KPIs (Readmission Rate, LOS, and Cost and implement a "Community Health Navigator" for repeat attenders with 3 or more visits within 90 days.

Chief Financial Officer (CFO): Financial Risk & Payer Analysis

- **Revenue Leakage: Uninsured patients** represent a high financial risk, exhibiting the highest readmission rate at **77.1%**.
- **Payer Gaps:** There is a significant gap between the **Total Claim Cost** and **Payer Coverage**. Medicare and Blue Cross Blue Shield patients show high average claim costs, leaving substantial "cost post-payer coverage" that affects the bottom line.
- **High-Cost Drivers: Urgent Care** clinic encounters result in unexpectedly high costs (\$23.3M total) despite zero recorded overnight stays, indicating high resource intensity per visit.

Recommendation:

Invest in a transition program to help eligible patients enroll in State Health Exchanges during their first visits to reduce uncompensated care losses.

Chief Medical Officer (CMO):

High-Risk Departments: Urgent Care (74%) and Emergency (64%) departments record the highest cases of re-admissions.

- **Procedure-Based Costs:** The procedure incurring the most cost post-insurance coverage is **Electrical cardioversion**, followed by **Fetal heart auscultation** and **Uterine fundal height evaluation**.
- **Readmission by Diagnosis:** Patients with chronic conditions and acute cardiac issues show the strongest correlation with 30-day readmissions, suggesting a need for better post-discharge community care.

Recommendation:

Develop a specialized geriatric emergency track to manage the complex needs of elderly patients.

Operations Manager: Capacity & Discharge Planning

- **Departmental Bottlenecks:** The **Inpatient** department is the primary bottleneck with an average stay of **36.8 hours**.
- **Efficiency Gaps:** The **Outpatient** and **Ambulatory** departments show high average Length of Stay.
- **Bed Occupancy:** Emergency encounters (2,322 total) has a patient return rate of 65.1%.

Recommendation:

Map the end-to-end discharge process for Inpatient and Emergency departments to identify and remove the largest bottlenecks in bed turnover

3. Summary

The recommendations focus on mitigating the 60.18% readmission rate through clinical specialization, improving financial stability by enrolling uninsured patients in coverage programs, and optimizing operational flow by addressing Inpatient department bottlenecks. These actions aim to align hospital performance with national benchmarks while reducing revenue leakage.

4. Github Portfolio Documentation

GitHub Link: <https://github.com/ChimchetamIbeh>

My portfolio features a comprehensive healthcare operational analysis of Massachusetts General Hospital. It contains structured datasets (27k+ records), automated project reports, and interactive dashboards that identify critical trends in patient re-admissions and departmental cost structures (\$36M+).